Sequence Learning Experiment
PSY310 Lab in Psychology
Lab Report

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GitHub Link:
https://github.com/tamannaz06/Tamanna-Chandna_Sequence-Learning

Introduction

Sequential learning is the process through which individuals gain understanding about the order and patterns in which events, stimuli or actions occur. It essentially involves recognising regularities and relationships between successive events, allowing them to be able to anticipate what is likely to happen next.

It's important to note that sequential learning is most of the times implicit in nature, meaning that individuals can understand an underlying sequence without consciously intending to learn it or being aware of the patterns they have learned. One of the most commonly used methods to study this process is the Serial Reaction Time (SRT) task. In this participants respond to stimuli appearing in different locations on a screen. With repeated exposure to a particular structured sequence, participants typically have faster reaction times, suggesting that they have understood the sequence even when they cannot consciously describe it. This establishes the successful relationship between learning and contingency (if-then effect). Willingham et al. (2000) further studied implicit motor sequence learning and showed that learned sequence knowledge is represented in response locations. Thus, sequential learning highlights how individuals develop sensitivity to patterns through repeated experience, even without conscious awareness of the learning process (Schwarb & Schumacher, 2012).

Method

Participants

This experiment involves data from five undergraduate students (n=5) from Ahmedabad University. The age ranges from 18-22 (mean=19.8). Both male and females took part: 3 males and 2 females. The demographics are given in the table below. The participants were acquired through convenience sampling: the experimenting laptop was taken to the University Cafeteria and given to people who were available for 15-20 minutes. It was made sure that there is no one distracting them while they do the experiment. 3 of them completed the task later in the evening after agreeing to participate. Basic instructions were made clear but the details about the trials shifting from a sequential order to a random one was not told. Consent was taken from all participants and their data is kept confidential by assigning it to a number and not their names.

Assigned no.	Age	Sex	Year of study
#1	20	Male	3rd
#2	20	Male	3rd
#3	18	Female	1st
#4	19	Female	2nd

#5	22	Male	4th

Table 1: Details of the participants

Materials

The experiment was conducted on a laptop with PsychoPy Studio 2026.2.4 installed. The laptop has a monitor of dimension 13.3”, a resolution of 1920x1080 pixels and a refresh rate of 60 Hz. 400 trials of the experiment were conducted. The aim was to see the difference in Reaction time between sequential and random conditions.

Stimuli and Procedure

The experiment begins with an ‘instructions’ routine that explains the task to the participants. Each trial began with a fixation cross presented at the centre of the screen for 1000ms (1 second). After the fixation, four vertical rectangles appear in a horizontal row, equally spaced from each other at the center of the screen. The x positions for each of these rectangles were fixed at -150, -50, 50, 150 pixels respectively and they were sized 100 by 10. These 4 rectangles were presented for 1 second after which a small triangular probe appeared above one of the rectangles. It remained there until a keyboard response was made irrespective of whether or not it was correct. Then the trial progressed with the presentation of the next fixation cross and so on. The first 200 trials (nreps=50) have a particular pattern of responses while the second half that is 200-400 have randomized probe placements.

Figures 1-3 show the stimuli and the sequence of the experiment.

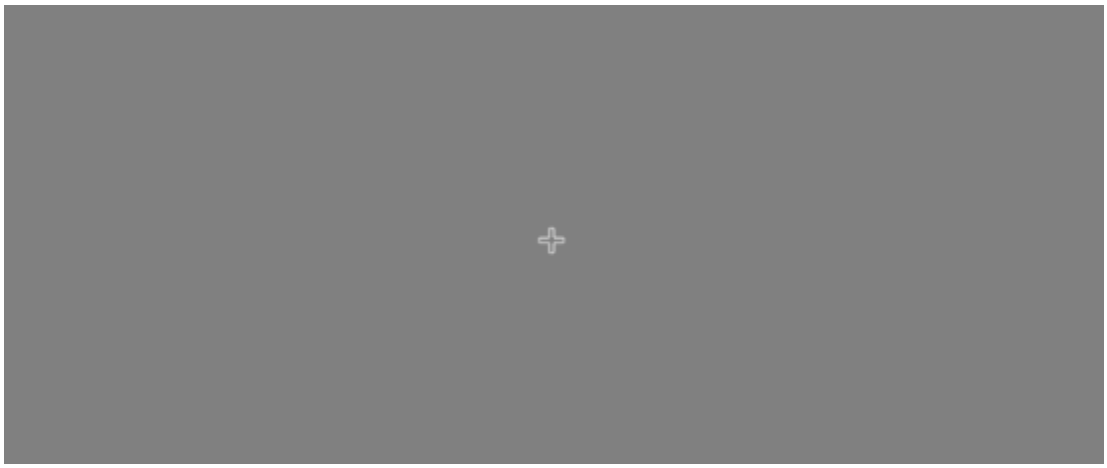


Figure 1: Fixation cross presented for 1 second before each trial.



Figure 2: Four vertical rectangles presented for 1 second before the triangle probe.



Figure 3: Four vertical rectangles with a probe above the (-50,0) rectangle.

The position of the triangular probes over specific rectangles corresponded with a sequence of 4 keys on the keyboard. These were letters: Z, X, C and V. Participants' task was to respond with the correct key as quickly as possible. The accurate responses were as follows: -150: Z, -50: X, 50: C, 150: V. The correct responses and conditions were coded in an excel table and connected to PsychoPy. Table 2 shows this table.

Position	Correct Keyboard response
-50	x
150	v
50	c
-150	z

This order of -50, 150, 50 and 150 (responses x,v,c,z respectively) was the ordered or sequential responses shown in the first half of the experiment, that is until 200 trials.
In the second half, that is, random condition, PsychoPy was coded to present the probe on any location with an equal probability of any location appearing next. Important data from the trials which included accuracy and reaction times were recorded by PsychoPy in a .csv file for each participant and were then analyzed as seen in the results section ahead.

Results

The first stage of analysis involved removing all the inaccurate trials and outliers from the data. In the beginning 2-3 trials most participants had higher reaction time as they were getting a hang of it, hence, such outliers were removed. In fact, the third participant had a reaction time of 19 seconds for the first trial as she was confirming the keys. Sometimes, due to the length of the experiment and extraneous factors, the participants simply get distracted in between and thus take longer to respond. It's important to remove such trials from the analysis as it skews the results. The trials with Reaction time over 2 seconds have been omitted. On average, 384.2 trials were valid out of 400.

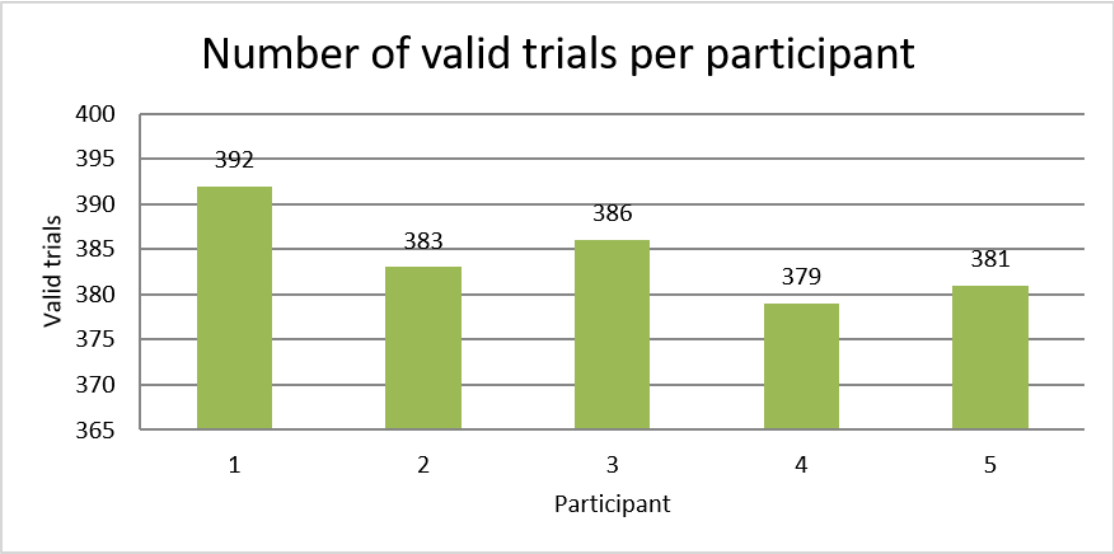


Figure 4: Number of valid trials per participant (inaccurate trials and outliers removed)

Next, the means for both sequential and random trials were calculated using only valid trials, for each participant. The results are presented in figure 5.

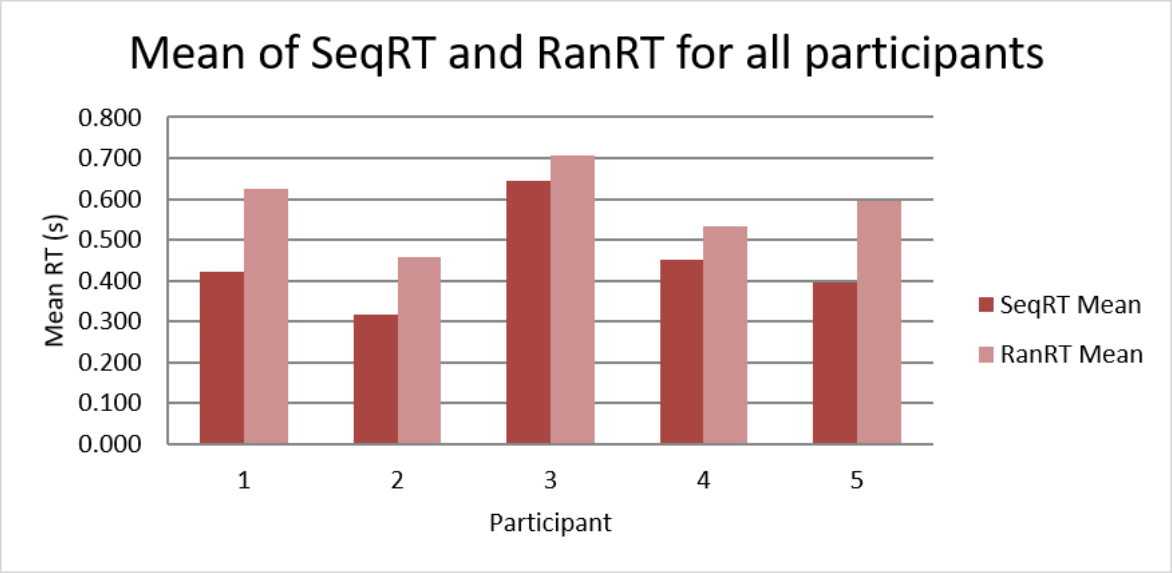


Figure 5: The averages of valid sequential and random conditions for all participants

For all participants randomized trials had a greater reaction time.

The following table shows the standard deviations for both the types of conditions for each participant.

Participant	SeqRT STD DEV	RanRT STD DEV
1	0.195	0.297
2	0.183	0.102
3	0.308	0.207
4	0.243	0.180
5	0.196	0.183

Table 3: Standard Deviations of valid trials for random and sequential conditions

Now, for each participant, their RT across 400 trials is plotted on a scatter plot line graph and their performance in the two conditions (sequential and random) is plotted in a combined line graph in order to compare the differences in their performance in both the conditions. Again, only valid trials are used for graphical data representation.

Participant 1

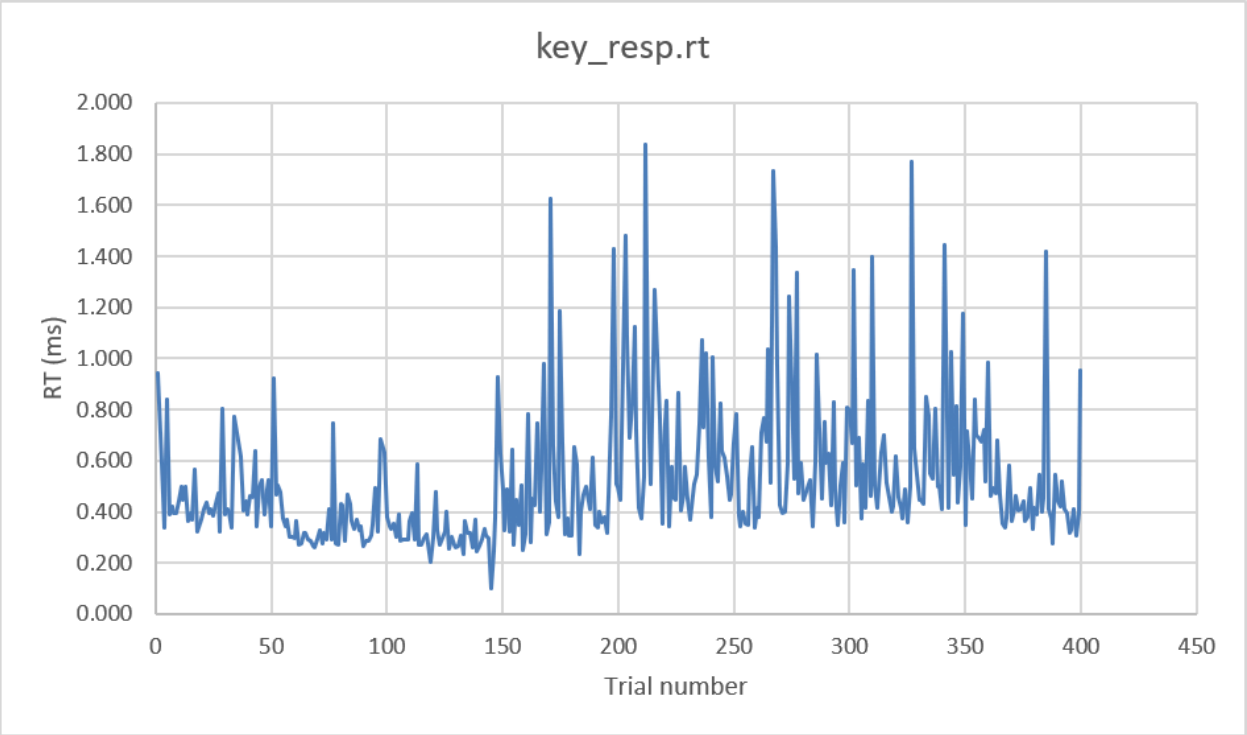


Figure 6: Graph representing participant 01's RT performance across 400 trials

This participant's performance clearly changed at the 200th trial as seen in the graph. The average RT became higher after 200 trials.

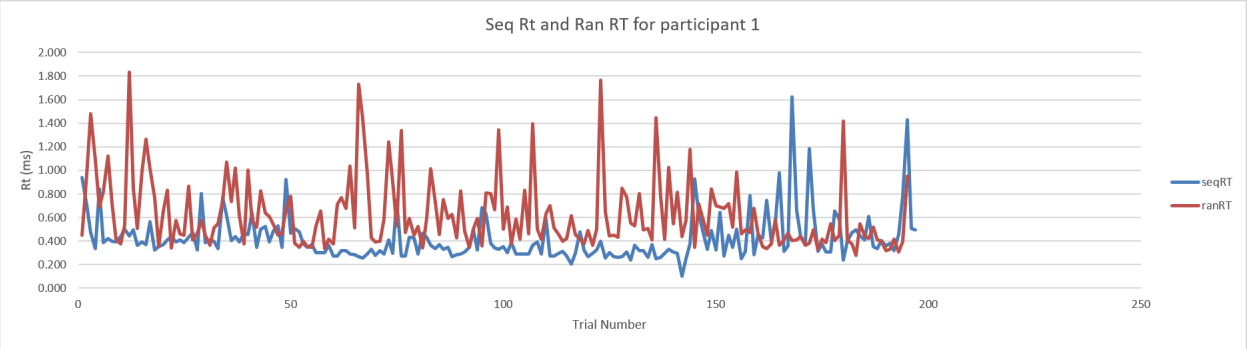


Figure 7:Graph comparing participant 01's RT performance across the sequential and random trials

The graph above shows that participant 01's performance was on average much faster on average in the sequential trials than in the random trials. The difference between the two means of the Sequential and random trials is 0.202 seconds.

Participant 2

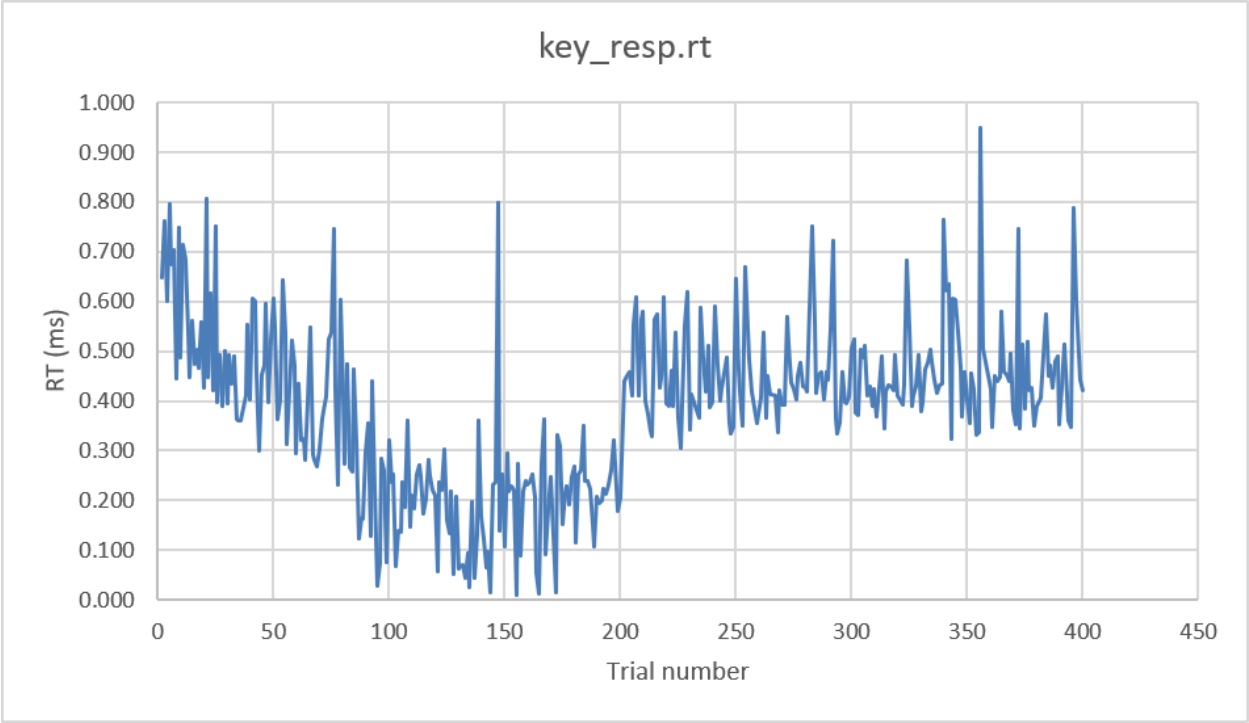


Figure 8: Graph representing participant 02's RT performance across 400 trials

This participant demonstrates the hypothesis perfectly as the RT reduces considerably once the pattern is learned and increases when it changes from sequential to random condition.

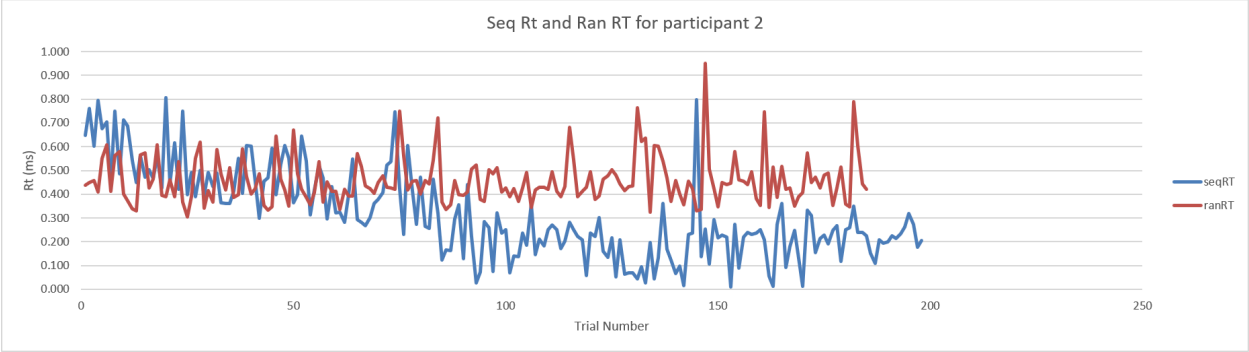


Figure 9:Graph comparing participant 02's RT performance across the sequential and random trials

The graph above shows that participant 02's performance was on average much faster in the sequential trials than in the random trials especially in the second half of both the routines. The difference between the two means of the sequential and random trials is 0.139 seconds.

Participant 3

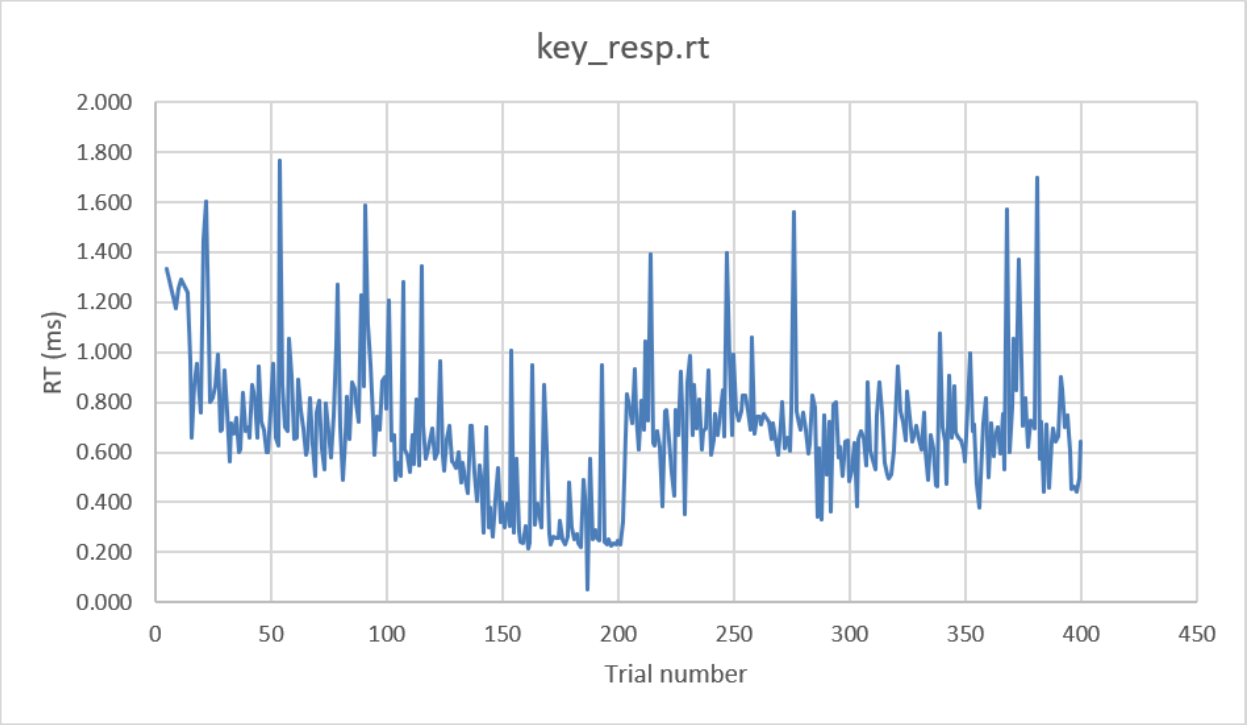


Figure 10: Graph representing participant 03's RT performance across 400 trials

This participant also shows increased RT after the conditions change from sequential to random. Also, the RT in sequential trials was higher in the beginning showing that the pattern was learned after doing about 100 trials.

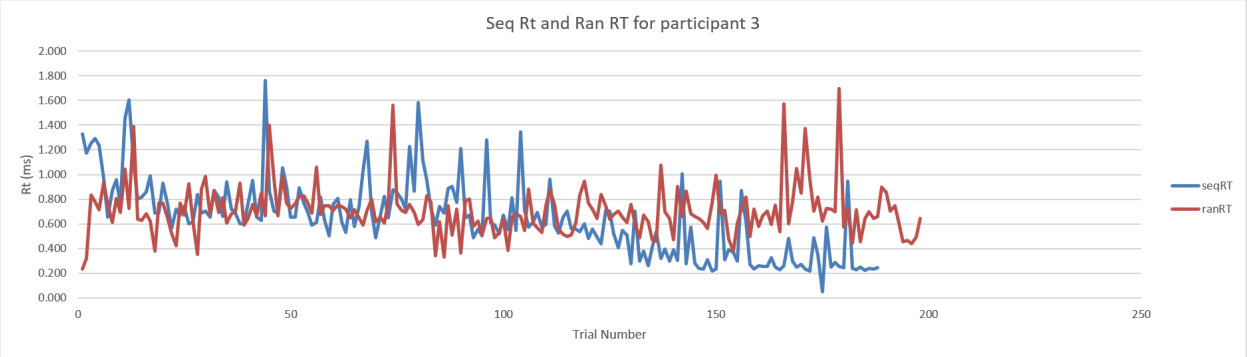


Figure 11:Graph comparing participant 03's RT performance across the sequential and random trials

The graph above shows that participant 03's performance was mostly faster in sequential trials except in a small proportion wherein she might have been tired or distracted. The difference between the two means of the sequential and random trials is 0.60 seconds.

Participant 4

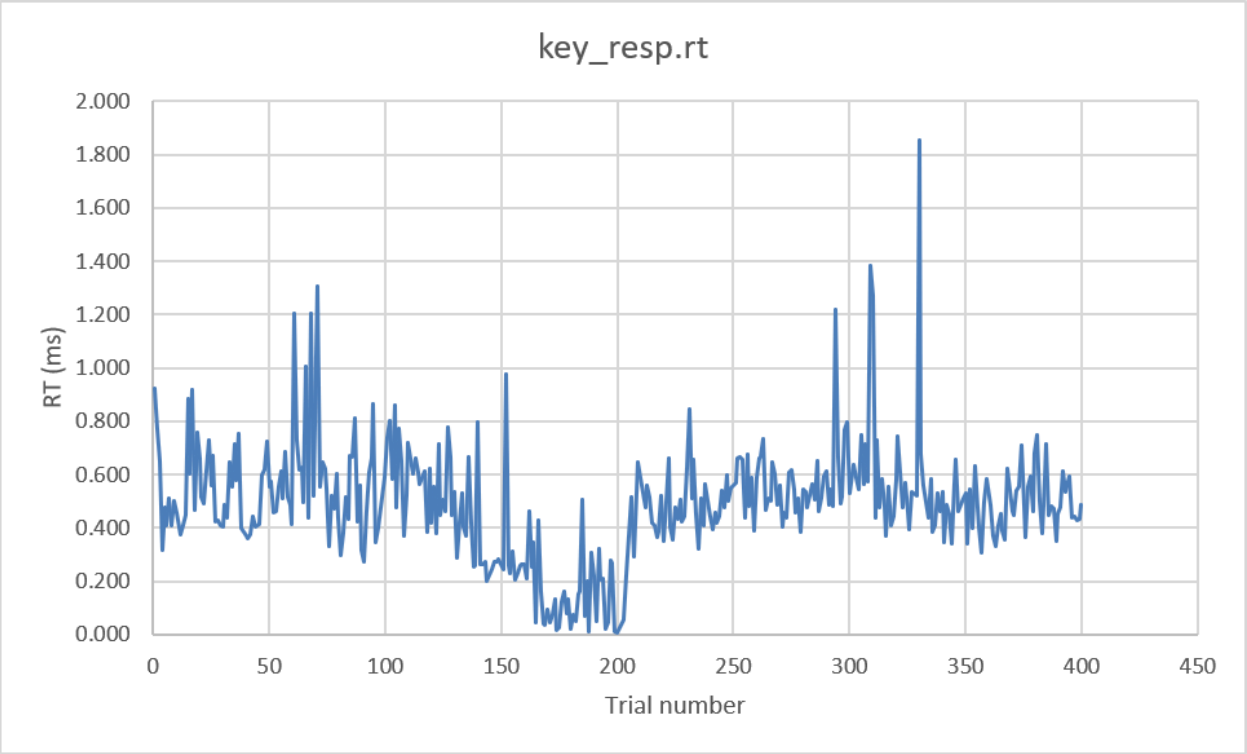


Figure 12: Graph representing participant 04's RT performance across 400 trials

Here, we can clearly see when the pattern was learned as the RT becomes much faster then. Once the conditions change to randomized order the RT becomes slower. This is very evident right after 200 trials.

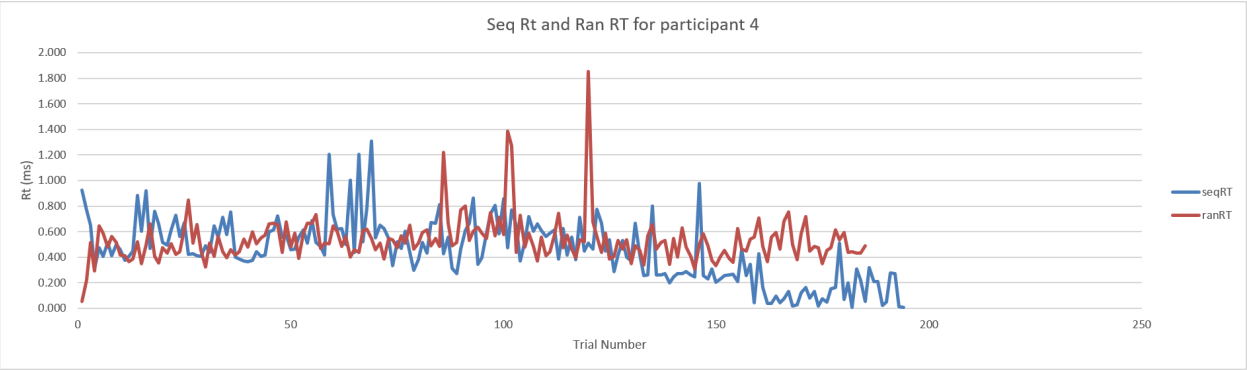


Figure 13:Graph comparing participant 04's RT performance across the sequential and random trials

The graph above shows that participant 04's performance was on average faster in the sequential trials than in the random trials except the first few routines wherein the pattern wasn't learned so the RT was similar for both. The difference between the two means of the sequential and random trials is 0.081 seconds which is comparatively not much.

Participant 5

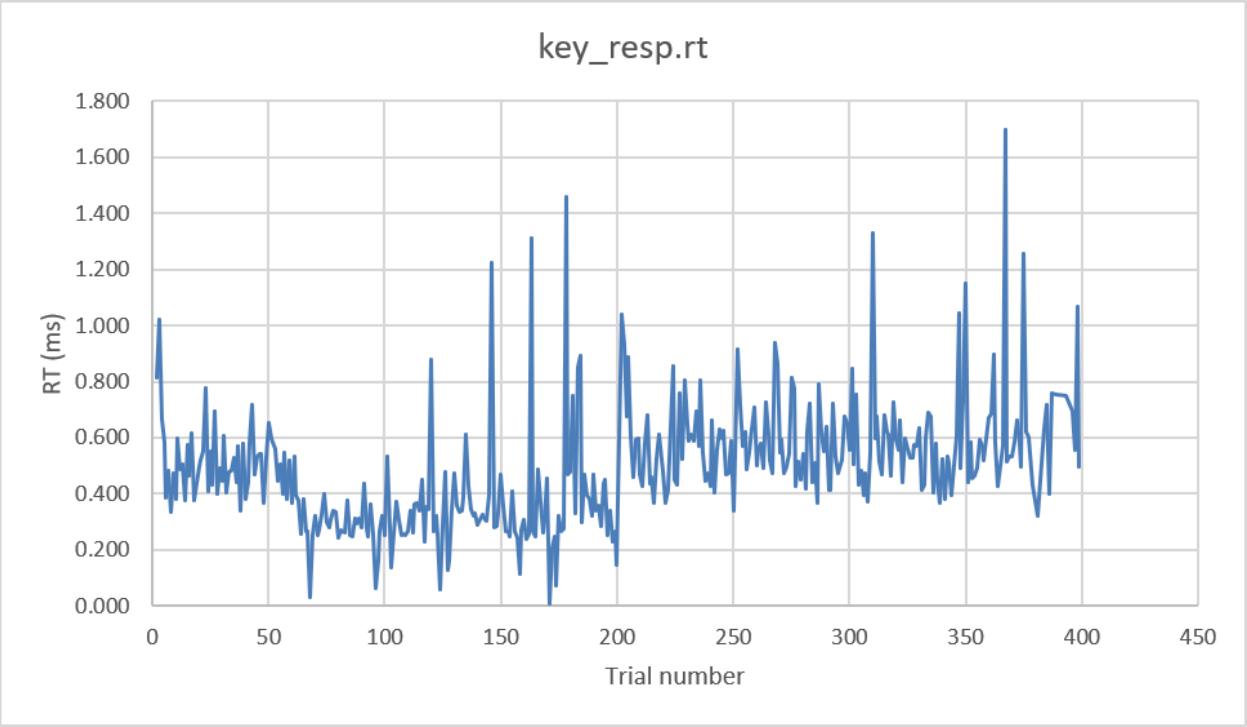


Figure 14: Graph representing participant 05's RT performance across 400 trials

Participant 4 found the pattern faster than the others after around 50 trials itself. He seems to have been slightly distracted in between but overall the RT for randomized conditions is much slower.

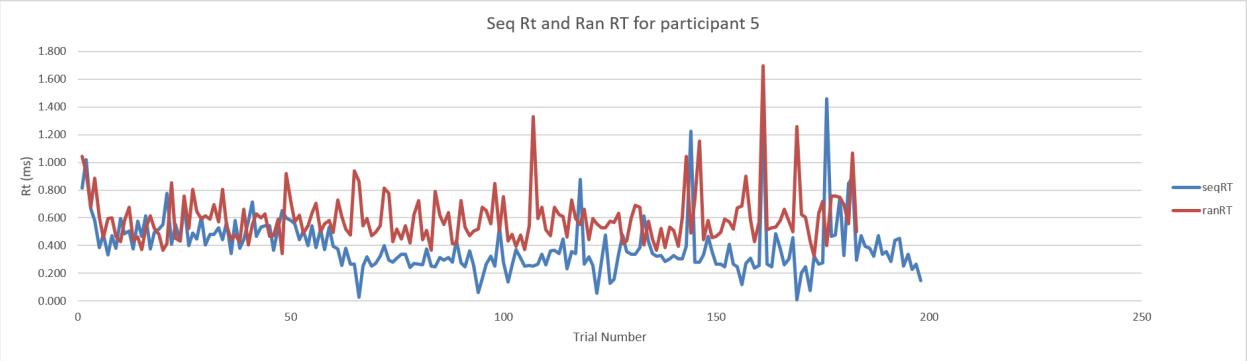


Figure 15:Graph comparing participant 05's RT performance across the sequential and random trials

The graph above shows that participant 05's performance was on average much faster in the sequential trials than in the random trials after the initial few routines. The difference between the two means of the sequential and random trials is 0.201 seconds.

Discussion

The results are consistent with the hypothesis that contingencies were learned in the sequential task. All five participants had a lower mean RT in the sequential conditions than in the random one. The overall mean RT (across participants) differed between the two by approximately 0.14 s, with the sequential section being around 0.45 s and the random section being around 0.58 s. Because key frequency and the motor demands were matched, the speed advantage is best explained by participants picking up the order.

There was also clear evidence of anticipatory responses as many sequential responses were very rapid, with some being less than 0.1 s. For instance, participant 2, had 22 consecutive correct responses under 0.1 seconds. RT this fast means that the participant pressed the next key before the probe could be processed. They were reacting to what they should expect, rather than what they saw. This is consistent with the theory of combining perception and action to predict the next items in a sequence proposed by Abrahamse et al. (2010).

The effects were of varying sizes in different individuals. The differences were quite large for participants 1 and 5 (about 0.20 s), moderate for participant 2 (about 0.14 s) and only small for participants 3 and 4 (about 0.06 s and 0.08 s respectively).

For this study all subjects performed the sequential section first (trials 1-200) then the random trials second (trials 201-400). This has the possibility of making the RT difference confounded with conditions because the order is the same for all participants. The second section would be faster if practiced. It will slow down if the participants are bored or pay less attention. Also, the random condition followed immediately after 200 trials of the learned sequence. Therefore, hopes are formed. In fact, one participant asked the experimenter if the following trials won't have any pattern, essentially trying to guess the experimental aim. This means that part of that difference may be due to the cost and not benefit, of the sequence being taken off, rather than being learned. With counterbalancing, we're able to compare both orders directly which is a much more convincing evidence of learning of the sequence.

References

Willingham, D.B., Wells, L.A., Farrell, J.M. et al. Implicit motor sequence learning is represented in response locations. *Memory & Cognition* 28, 366–375 (2000). https://link.springer.com/article/10.3758/BF03198552

Schwarb H, Schumacher EH. Generalized lessons about sequence learning from the study of the serial reaction time task. *Advances in Cognitive Psychology, 8*(2), 165–178 https://pmc.ncbi.nlm.nih.gov/articles/PMC3376886/

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